

Space Planning Automation

Architecture recommendation

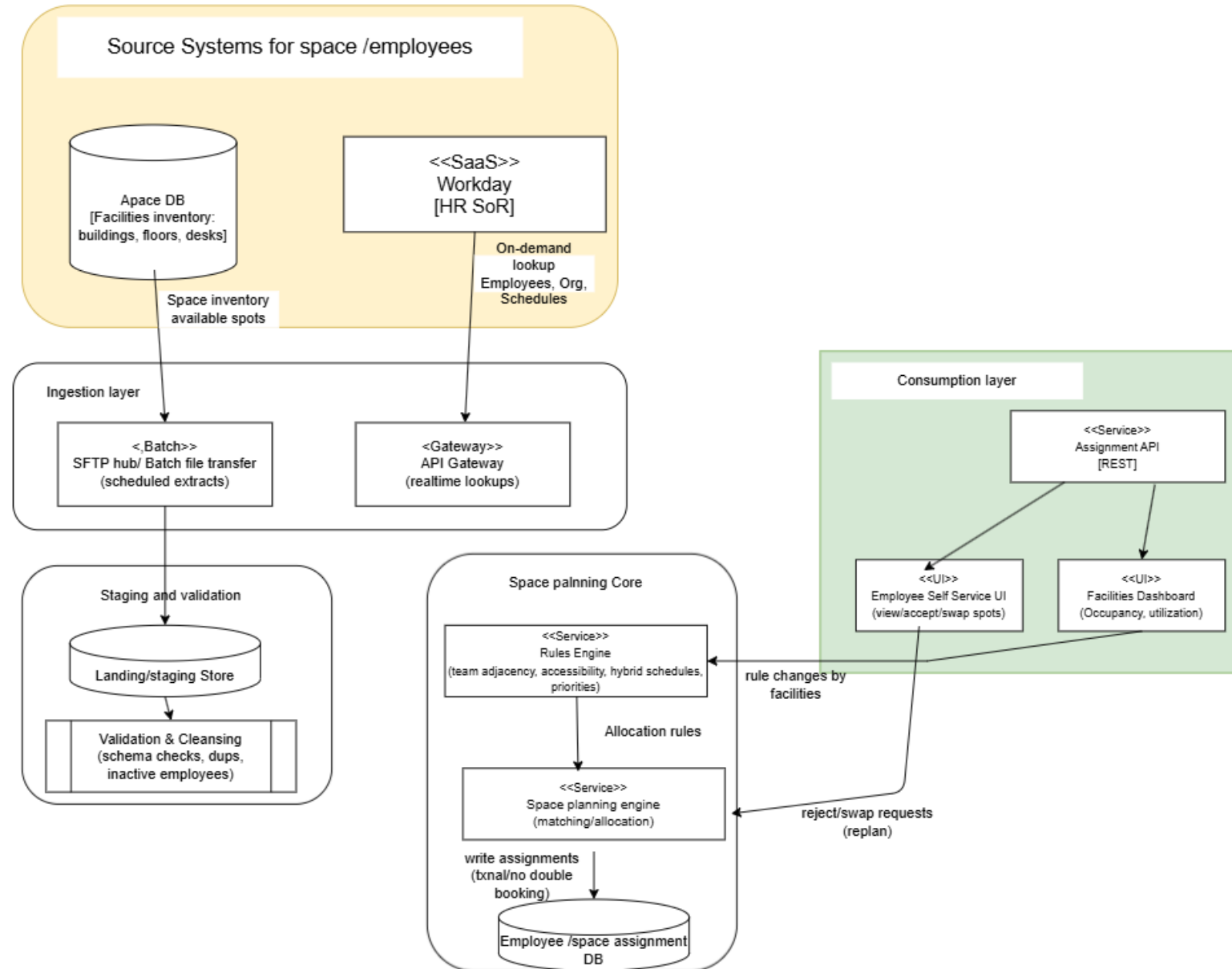
Problem Statement

Employee data (Workday) and space inventory (facilities DB) exist in separate systems on off-prem cloud with no automated way to match people to available workspaces. The organization needs a space planning system that consolidates both sources, automatically assigns employees to suitable spots based on configurable rules (team adjacency, accessibility, hybrid schedules), prevents double-booking, and provides employees a self-service view of their assignment and facilities teams a view of utilization.

Architecture PoV and assumptions

- Workday remains the system of record for employee data (identity, org structure, work location, employment status); the space planning system never modifies HR data, only consumes it.
- The Space DB (facilities inventory) is maintained by the facilities team and is accurate and current buildings, floors, desks, and their attributes (accessibility, equipment, capacity) are kept up to date at source.
- Enterprise integration infrastructure (API gateway, SFTP hub) already exists and is available for this solution; no new integration platform is being procured.
- Authentication and authorization federate (ADFS) through the enterprise identity provider; employees see only their own assignment, while facilities roles have broader visibility (RBAC).
- Nightly batch synchronization is sufficient for the baseline planning cycle; real-time API lookups are reserved for exception cases (new hires, transfers, same-day changes).
- Allocation rules (team adjacency, accessibility needs, hybrid schedules, priority tiers) are defined and owned by the facilities/workplace team, and can change without code releases via the rules engine.
- Data privacy requirements are met by consuming only the minimum employee attributes needed for planning - no compensation, performance, or other sensitive HR data leaves Workday.
- Initial phase covers desk/space assignment only; meeting-room booking, visitor management, and badge-system integration are out of scope for phase 1.

Component Diagram



Availability

- The planning engine is stateless and horizontally scalable; multiple instances run behind a load balancer, so no single instance failure interrupts service.
- The assignment database is the availability-critical component and runs with replication and automated failover; the Assignment API serves reads from replicas if the primary is lost.
- Batch-based design is inherently resilient to source outages: if Workday or the Space DB is unavailable, the system continues serving current assignments from its own store - an ingestion failure delays re-planning but never takes down lookups.
- A failed or partial nightly load never corrupts the active dataset: validation runs against staging, and the planner only switches to new data once a load is complete and verified (load-then-swap).
- Assignment writes are transactional with conflict detection, so a failover or retry can never produce a double-booked space.
- The self-service UI and dashboards depend only on the Assignment API, not on the sources - Workday downtime is invisible to end users.
- Recovery targets are modest and should be stated as such: this is a workplace convenience system, not a payment system - an RTO of a few hours and RPO of one batch cycle (24h) is acceptable and keeps the DR footprint cheap.